

- 2 The risk of developing cardiovascular disease (CVD) can be increased by a number of factors including diet and a history of thrombosis.

Thrombosis is the development of a blood clot in a blood vessel. Blood clots can develop in veins and then move through the circulatory system into the coronary artery.

- (a) A diet high in saturated triglycerides raises the levels of cholesterol in the blood.

(i) What is the ratio of glycerol to fatty acid in a triglyceride molecule?

(1)

- ☐ **A** 1:1
- ☐ **B** 1:3
- ☐ **C** 3:1
- ☐ **D** 3:3

(ii) Which diagram shows the bond that joins a glycerol molecule to a fatty acid in a triglyceride?

(1)

- ☐ **A** $\begin{array}{c} \text{H} \quad \text{O} \\ | \quad || \\ -\text{C}-\text{N}-\text{C}- \\ | \end{array}$
- ☐ **B** $\begin{array}{c} \text{H} \quad \text{OH} \\ | \quad | \\ -\text{C}-\text{N}-\text{C}- \\ | \end{array}$
- ☐ **C** $\begin{array}{c} \text{O} \\ || \\ -\text{C}-\text{O}-\text{C}- \\ | \end{array}$
- ☐ **D** $\begin{array}{c} \text{OH} \\ | \\ -\text{C}-\text{O}-\text{C}- \\ | \end{array}$



(iii) The table gives some information about four fatty acids.

Fatty acid	Number of double bonds between carbon atoms	Number of carbon atoms
butyric	0	4
stearic	0	18
palmitoleic	1	16
linoleic	2	18

Explain which of these fatty acids would have the lowest risk of causing CVD, if included in a diet in equal masses.

(2)

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(b) Anticoagulants, antiplatelets and thrombolytics are drugs used to treat blood clots.

(i) One anticoagulant binds to the active site of thrombin.

Explain how this drug reduces blood clotting.

(2)

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(ii) Molecules on the surface of platelets enable them to bind to other molecules.

One of the antiplatelet drugs affects molecules on the surface of platelets.

Explain how this drug reduces blood clotting.

(2)

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